Skylar: Troubleshooting Predictive Alerts

Guide	Skylar
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Background

The Predictive Alerts feature in Skylar is designed to enhance user experience by providing timely notifications and insights. However, users may occasionally encounter issues that hinder its functionality. This guide aims to equip you with the necessary steps and tips to effectively troubleshoot common problems associated with Predictive Alerts. Whether you're facing notification delays, inaccuracies, or other technical glitches, our comprehensive troubleshooting strategies will help you resolve these issues swiftly, ensuring that you can fully leverage the capabilities of Skylar's predictive technology.

I think the displayed plot is showing a bad prediction

This happens. Please post in the skylar-engineering-team chat the plot, the observed behavior, the desired behavior, and what the reasoning is. Making adjustments to handle new scenarios is relatively straight forward as there is a series of unit tests that run which also helps prevent regressions.

No prediction plot is shown

If no prediction plot is shown, or if there is a UI error related to displaying the prediction plot, please refer to this document: Unable to View Predictive Events

Algorithm Overview

There are some peculiarities for each type of metric, but in general predictive alerting is attempting to predict when a metric will reach 100% capacity. There are two components to this algorithm. First, linear regression is applied to the daily max values in an attempt to find the general daily growth rate. Additionally if the growth for today is a significant outlier in terms of typical daily growth, it will be assumed that something has changed and todays growth will be used as the “new normal.” There is then a simple calculation to determine how many days until the threshold will be reached.

Denoising Attempts

In order to reduce the number of false positives there are a few filters put in place.

- If it is predicted to reach threshold in more than 30 days.
- If it is predicted to reach threshold in less than 8 hours
- If the number of actual observations is too short. (We will forecast out at most three times the observation window.)
- If the current value is less than 33%
- If we have already issued an alert in the last 30 minutes.
- If there was not also an alert 6-8 hours ago
- Additional metric type specific denoising attempts (see below.)

Finding your Metric

Usually you will begin by finding your metric. First you need to know the device_id.

Memory and CPU

device_id is enough.

filesystems

You will also need filesystem name. (e.g. /tmp) and what is needed is the fs_id, which can be found in the reporting table fs_current.

```
SELECT * from fs_current where dvc_id = <your device id>
```

This will produce a list of all filesystems for that device_id. Find the row with value /tmp in column fs_nme. Then find the corresponding fs_id.

network interfaces

For network interfaces, you will need to look at the reporting table if_current.

```
SELECT * FROM if_current WHERE dvc_id = <your device id>
```

This will give a list of interfaces for that device_id; from this list we need to note the if_id for the interface you're looking for.

trouble shooting

What if fs_current or if_current is empty?

These tables are part of the reporting table schema, and can be a signal that reporting data is missing for some reason. Use this guide: Skylar: Troubleshooting Missing Reporting Data

Confirming Data

otel_metrics_gauge

This is where data flows into the anomaly detection / predictive alerting pipelines. Typically when debugging, you will just want to be sure that these tables are still ingesting new data, so the following query should be positive.

```
select count(*) from otel_metrics_gauge where TimeUnix > now() -
interval 15 minutes
```

aiml_metrics_obs

the aiml-anomaly-detection-clickhouse-poller pipeline will ingest data from otel_metrics_gauge and dump it here. Again, checking that this table is still producing new values is a good sanity check.

```
select count(*) from aiml_metrics_obs where obs_ts > now() - interval
15 minutes
```

Checking values for predictive alerting

It can be helpful to check data for the device in particular you're interested in.

filesystems

```
select obs_date , max(obs_val_f64) from aiml_metrics_obs
WHERE sl_dvc_id = 1
AND sl_fs_id = 1
AND metric_unit = 'Percentage'
AND obs_ts > NOW() - INTERVAL 30 DAY
GROUP BY obs_date
ORDER BY obs_date
```

network interfaces

For network interfaces, the metric_unit determines which aspect of the interface you're interested in: ifInOctets / ifOutOctets for incoming / outgoing traffic.

and then ifInErrors / ifOutErrors, ifInDiscards, ifOutDiscards – for issues with the interface. Note that for network interface alerts to rigger, we need errors/discards captured.

```
select obs_date , max(obs_val_f64) from aiml_metrics_obs
WHERE sl_dvc_id = 1
AND sl_if_id = 1
AND metric_unit = <metric_unit>
AND obs_ts > NOW() - INTERVAL 30 DAY
GROUP BY obs_date
ORDER BY obs_date
```

memory

```
select obs_date , max(obs_val_f64) from aiml_metrics_obs
```

```
WHERE sl_dvc_id = 1
AND metric_name = 'Physical Memory Utilization'
AND obs_ts > NOW() - INTERVAL 30 DAY
GROUP BY obs_date
ORDER BY obs_date
```

otel_logs

For memory, we also need otel_logs – alerts will only fire if there is a OOM log associated with high memory. Logs for your device can be found with the ResourceAttributes column in otel_logs.

CPU

```
select obs_date , max(obs_val_f64) from aiml_metrics_obs
WHERE sl_dvc_id = 1
AND metric_name = 'Overall CPU'
AND obs_ts > NOW() - INTERVAL 30 DAY
GROUP BY obs_date
ORDER BY obs_date
```

Checking Predictive Alert Tables

After data ingestion (otel_metrics_gauge → aiml_metrics_obs), which is handled by the clickhouse-poller service, the new data will be available for the predictive_alerting service. After running the algorithms to make a prediction, any metrics that have an alert will be inserted to the aiml_pa_q_in table, and then inserted into the aiml_pa table (unless it is a duplicate alert.)

The alerts in aiml_pa are then processed. The outcome here is that the alert will be published to the anomaly_alerts table for the UI to pick up or it will be filtered out. Either way, there should be an entry in aiml_pa_status indicating the outcome.

reasons for filtering

- Current value is less than 33%
- Prediction is too far out relative to the amount of data present (forecast window cannot exceed 3* the data window)
- Already alerted in previous 30 minutes
- No alert was generated 6-8 hours ago for the same series

trouble shooting

- The alert is not showing up in UI and it should.
  - First check the anomaly alerts table. If it's present here, but not in the UI then it is unlikely to be a predictive alerting issue. If it is not present, then check the aiml_pa_status table. If it is here, then the reason for filtering is given. If it is not present, then check the aiml_pa table. If it is here, then there was a pod crash between insertion into aiml_pa and aiml_pa_status. If it is not in aiml_pa, then either the algorithm did not alert or it did not run.
  - If there is a pod crash, the logs will show why and the most common reason lately has been table schema name changes or unable to reach the tenant service.
  - If you've checked the data flow, the alert tables, nothing is crashing, then it is likely because the algorithm did not alert. If you feel unsatisfied at this point, please contact `@David Daniels`

Important Pods

If you see these pods restarting a lot, that can clue in where the problem is with regard to predictive alerting. These are also good pods to check the logs of for any failures (that do not necessarily crash the pod.)

aiml-anomaly-detection-clickhouse-poller

This pod ingests data from otel_metrics_gauge and dumps to aiml_metrics_obs.

aiml-anomaly-detection-clickhouse-predictive-alerting

This pod runs the predictive alerting algorithm and publishes alerts to anomaly_alerts, which is then picked up by the UI.